

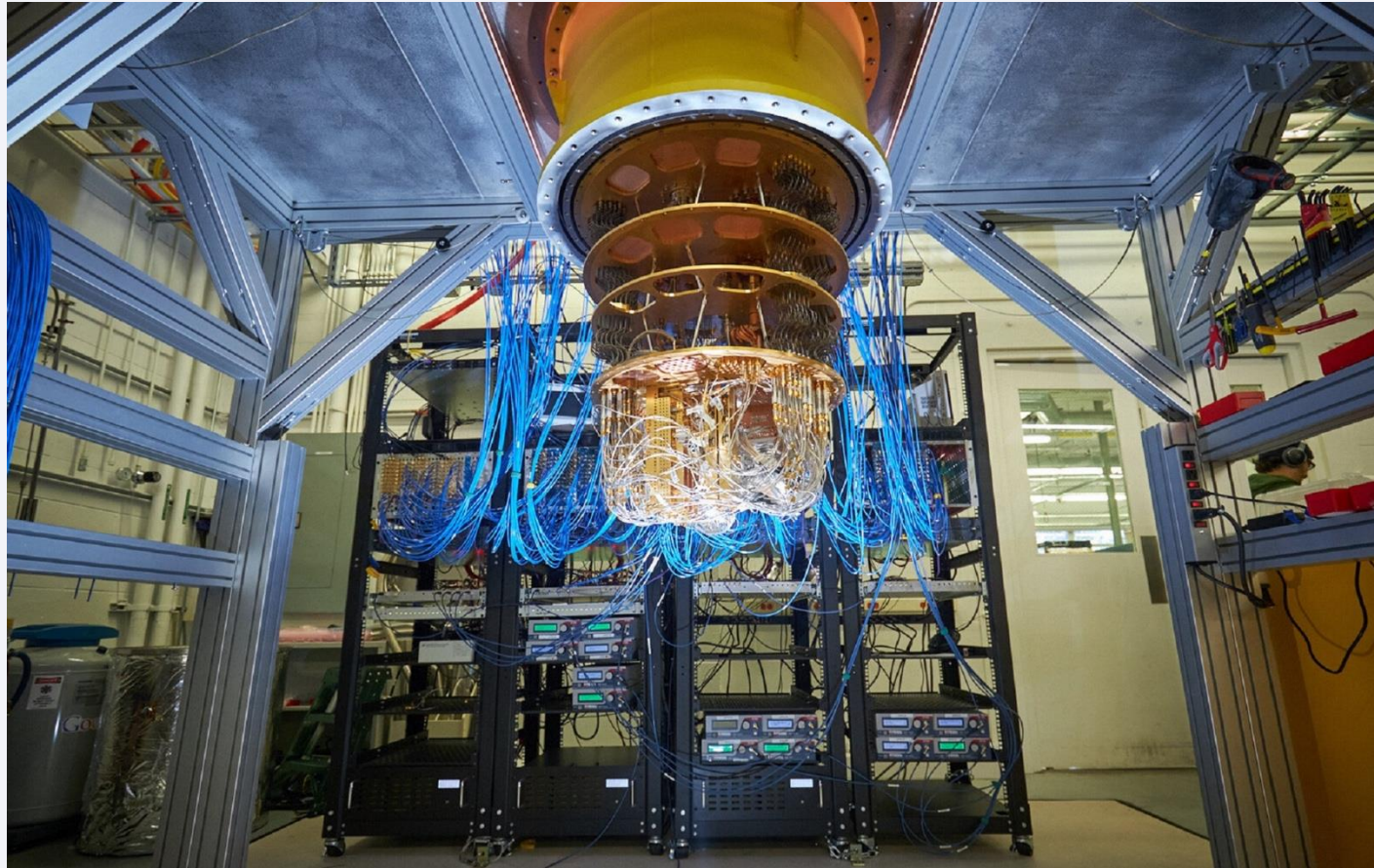


# Mapping the quantum limits of classical simulatability

L. Shackleton, Phys. Rev. Lett. 136, 186503 (2026)

L. Shackleton, in progress

# Simulating quantum systems is hard



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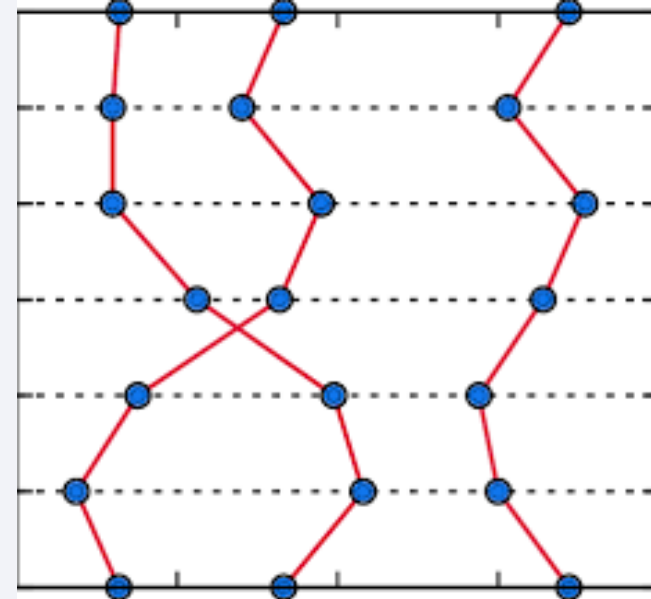
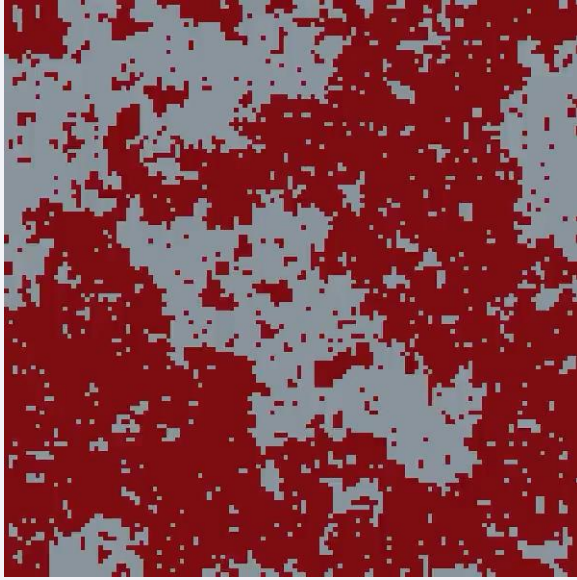
## Quantum Phase Transition at Nonzero Doping in a Random $t$ - $J$ Model

[Leyna Shackleton](#) <sup>1</sup>, [Alexander Wietek](#) <sup>2</sup>, [Antoine Georges](#) <sup>2,3,4,5</sup>, and [Subir Sachdev](#) <sup>1</sup>

The spectral function for the random  $t$ - $J$  model, calculated using the Lanczos algorithm on an 18-site cluster, is shown for several values of doping in Fig. 1. A



# Simulating classical systems is (more) easy



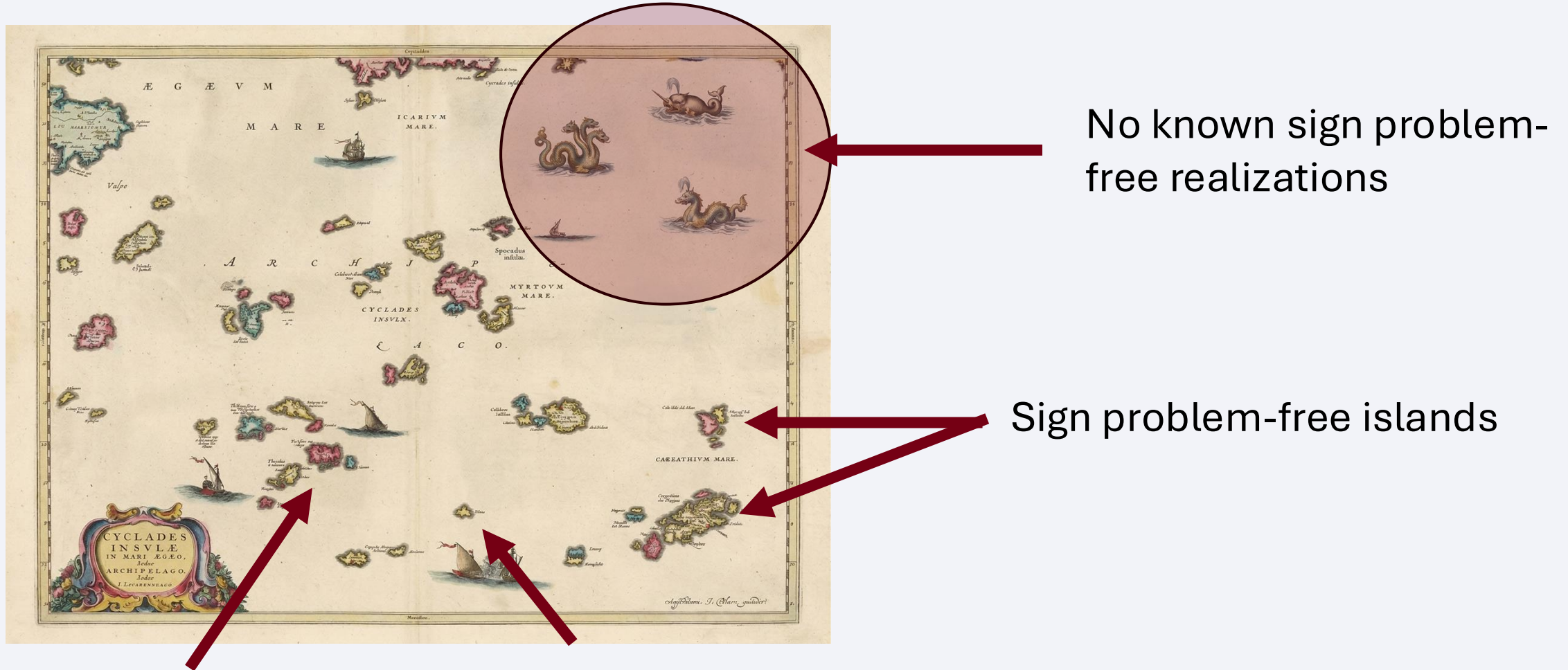
Sample via Monte Carlo

$$Z_{\text{classical}} = \sum_x e^{-E(x)/k_B T}$$

Complex weights

$$Z_{\text{quantum}} = \text{Tr} \left[ e^{-\hat{H}/k_B T} \right]$$

# The landscape of quantum phases



L. Shackleton and S. Sachdev,  
Phys. Rev. B **111**, 075101 (2025)

E. Berg, M. Metlitski, S. Sachdev,  
Science 338, 1606 (2012)

# Can a phase have an intrinsic sign problem?

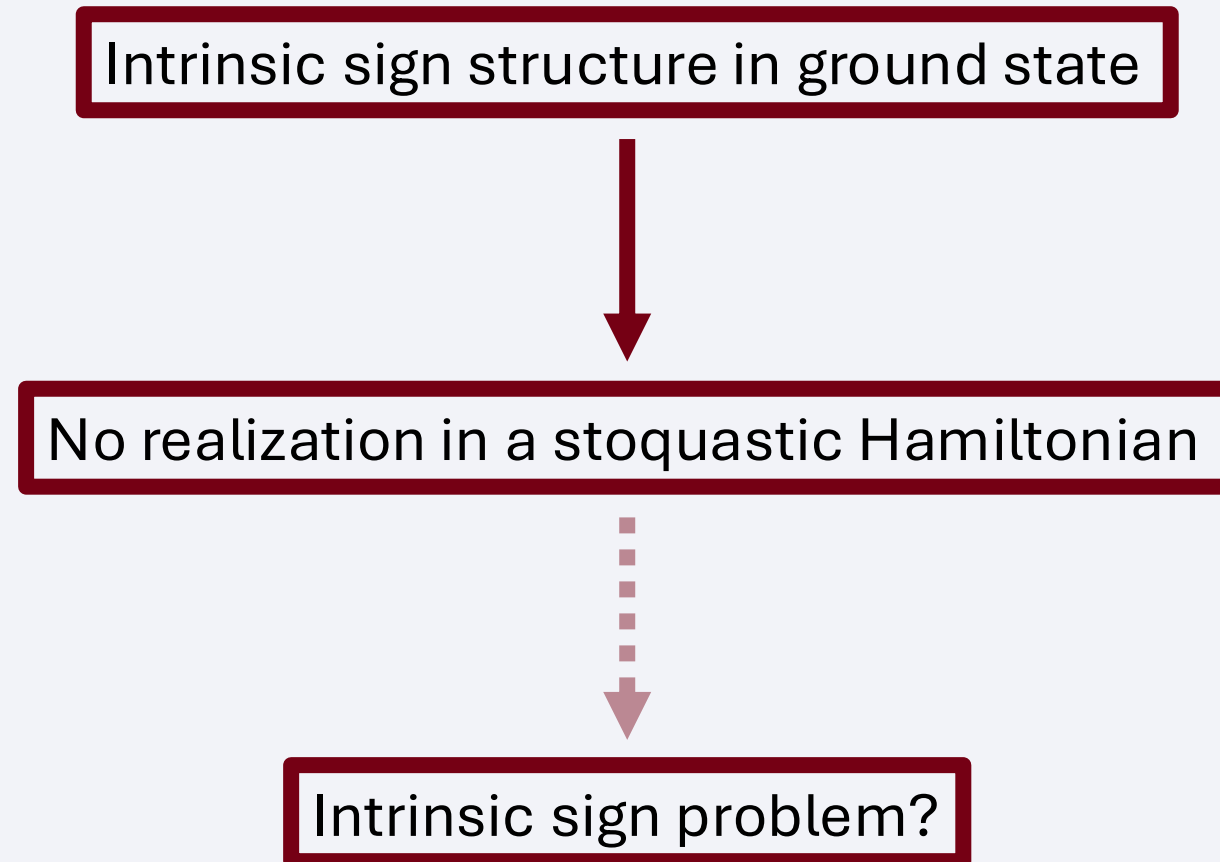
$$Z_{\text{quantum}} = \text{Tr} \left[ e^{-\hat{H}/k_B T} \right]$$



Completely positive matrix: “stoquastic”

Stoquastic Hamiltonians must have a completely positive ground state (Perron-Frobenius theorem)

# Ground state non-positivity as a proxy



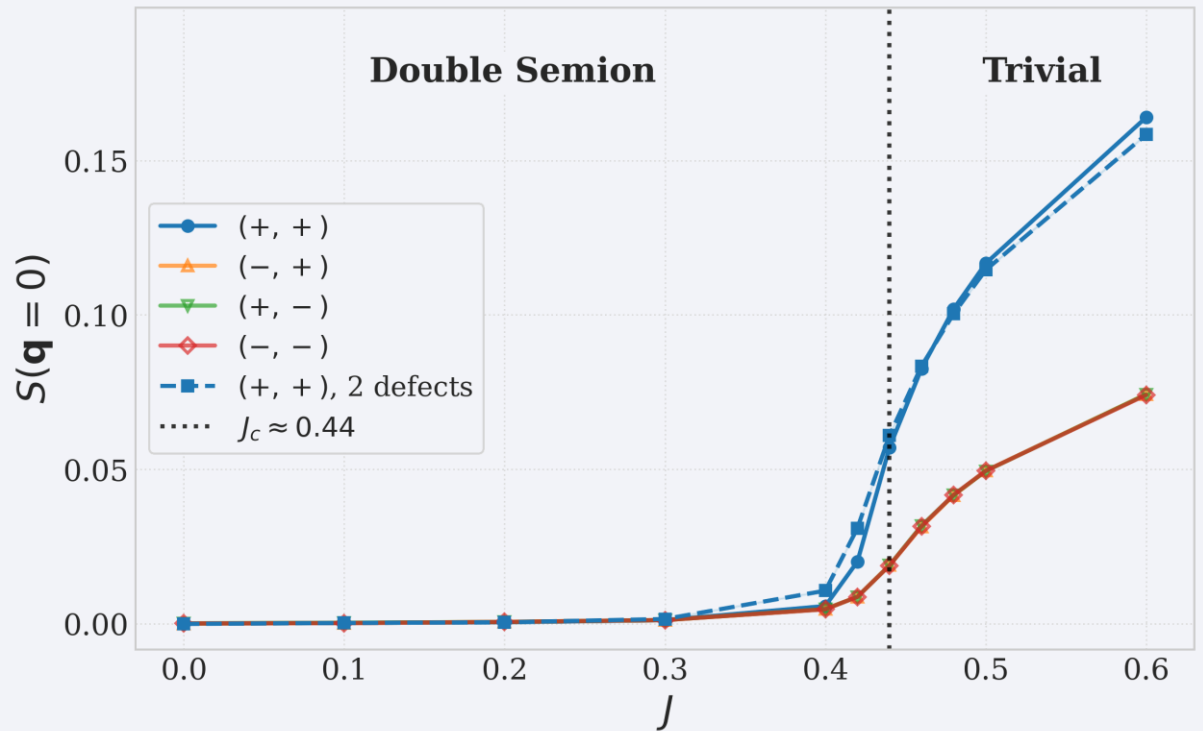
# No-go theorems for topological phases



- Double semion topological order (Hastings 2016)
- **Most** topological orders (Ringel 2017, Seo 2025)
- “*We analyze the Gauss sums of 405 different topological orders classified up to rank 12 [...] and find that 398 suffer from the intrinsic sign problem.*”

# Every twisted quantum double has a sign problem-free model

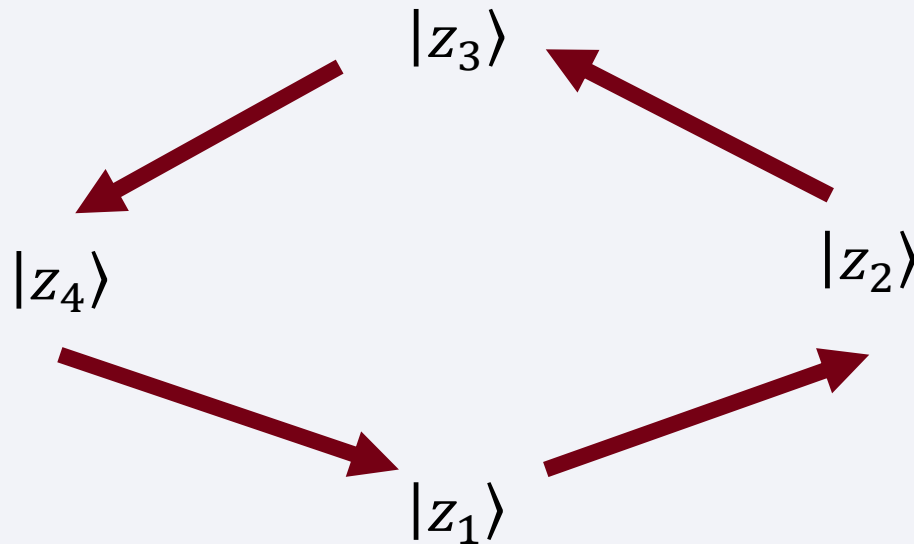
- Not fine-tuned
- No sophisticated tricks, uses conventional QMC methods
- Hamiltonians are standard and well-known



# Stochastic series expansion and the sign problem

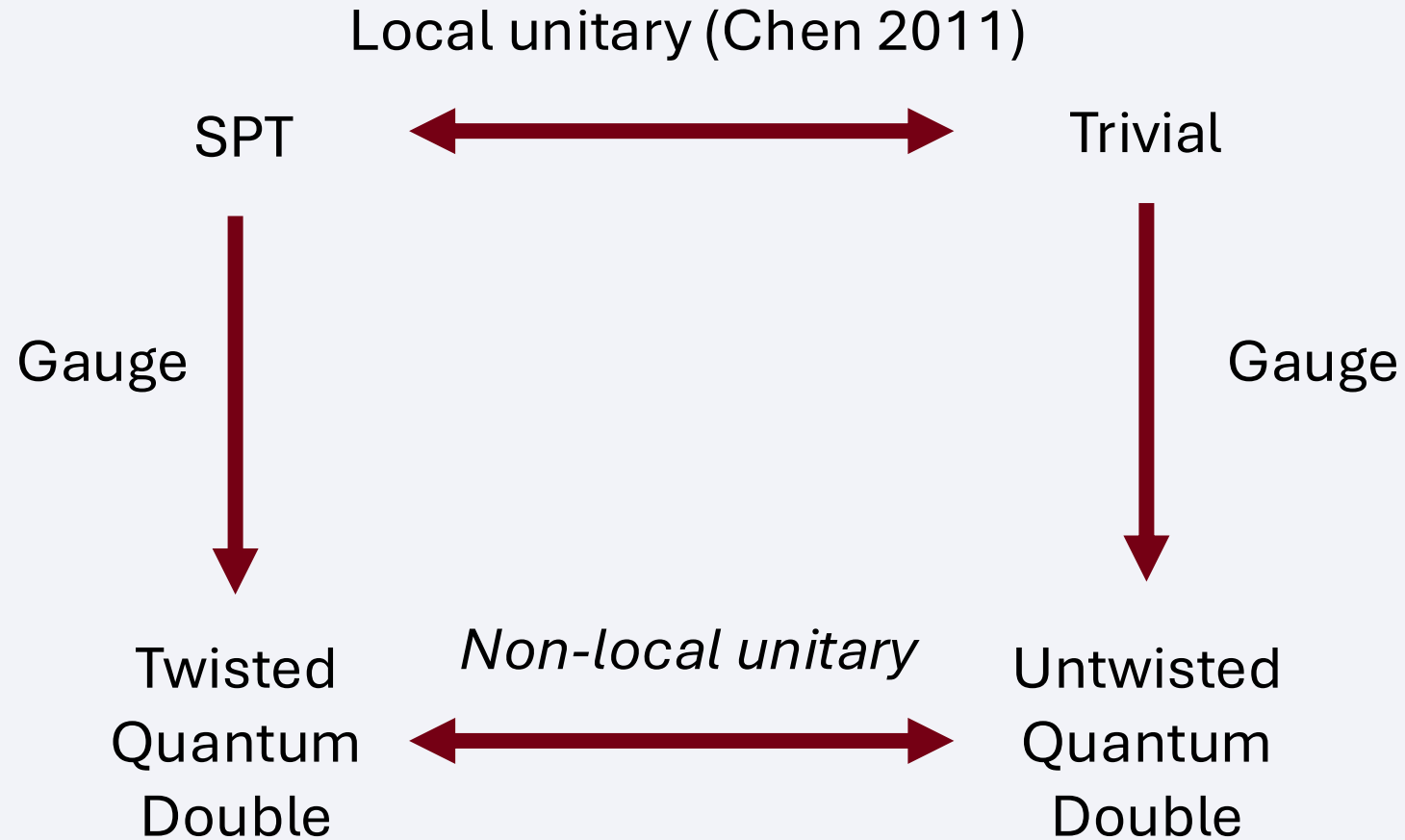
$$Z = \sum_j \sum_n \sum_{\{b_1, b_2, \dots, b_n\}} \frac{(-\beta)^n}{n!} \langle z_j | H_{b_n} \dots H_{b_2} H_{b_1} | z_j \rangle$$

$$H = \sum_b H_b$$



- Stoquasticity a sufficient but not necessary condition
- More general, global cancellation of phases
- Sign structure invariant under  $|z_j\rangle \rightarrow e^{i\theta_j}|z_j\rangle$
- Sign problem-free if *can be made stoquastic* via many-body phase rotations (Hen 2021)

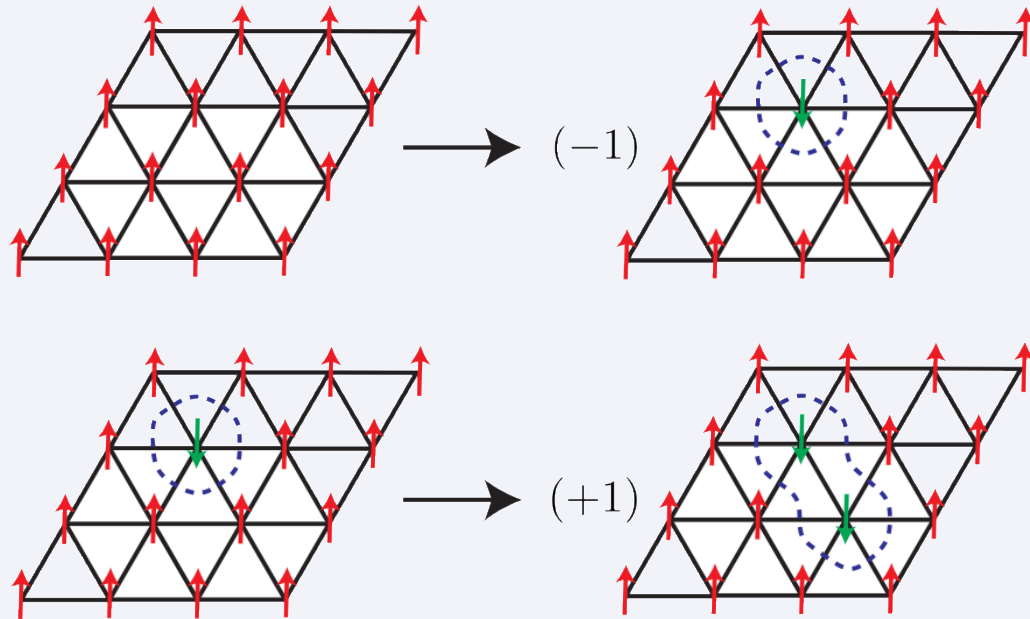
# Main idea



Ex: Levin-Gu SPT gauges to double semion

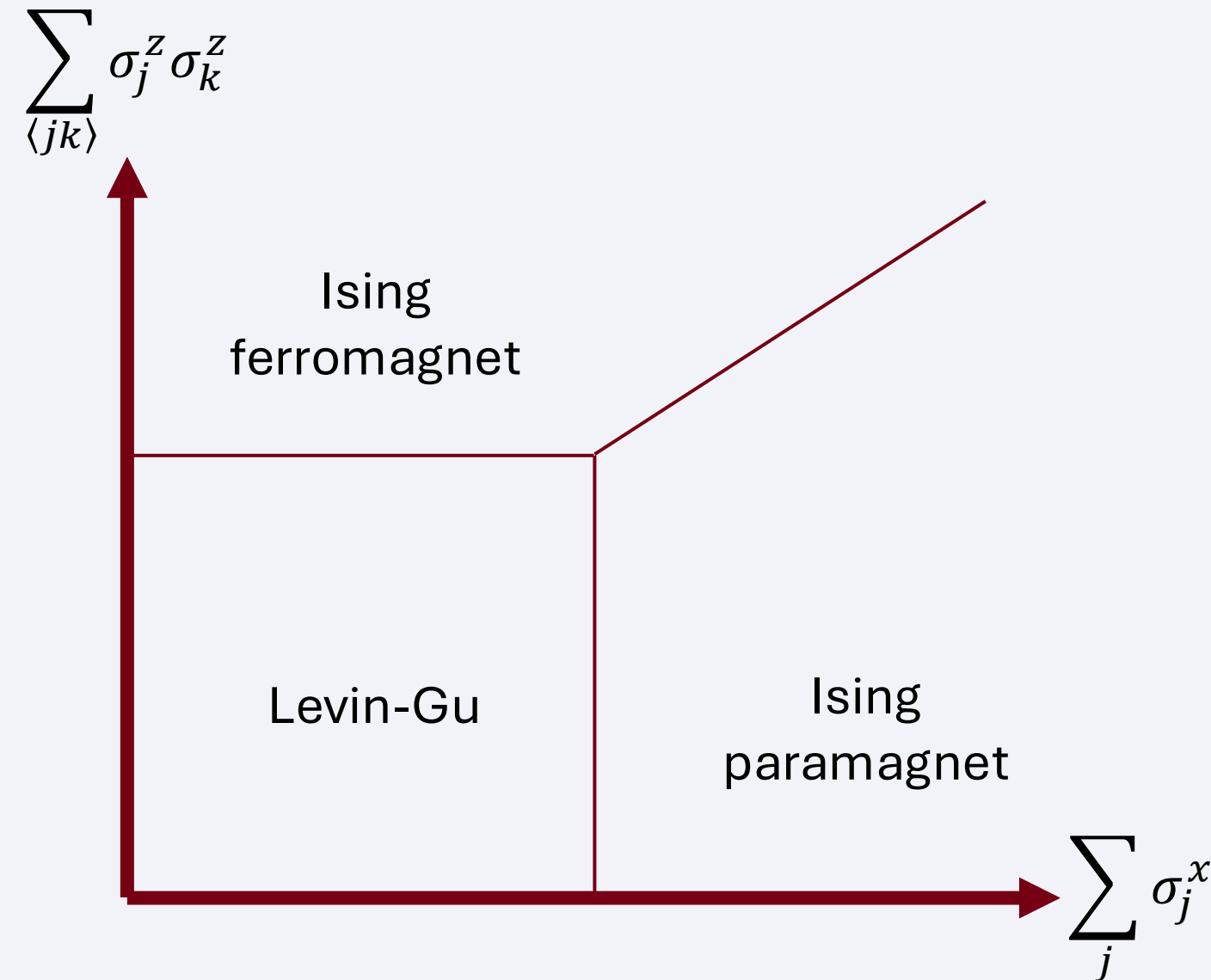
# The Levin-Gu model is sign problem-free

$$H_{\text{Levin-Gu}} = \sum_j \sigma_j^x \prod_{\triangle jkl} i^{\frac{1}{2}(-1 - \sigma_k^z \sigma_l^z)} \equiv - \sum_j B_j$$



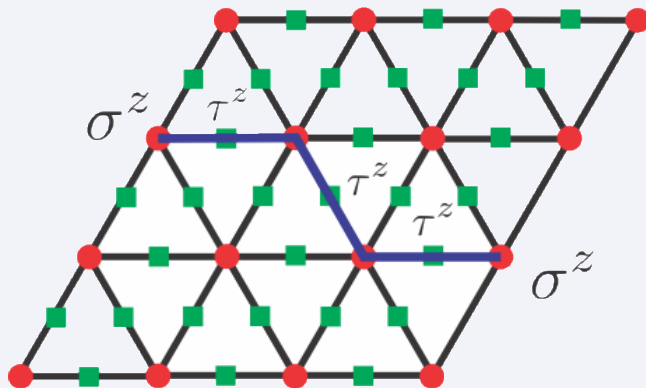
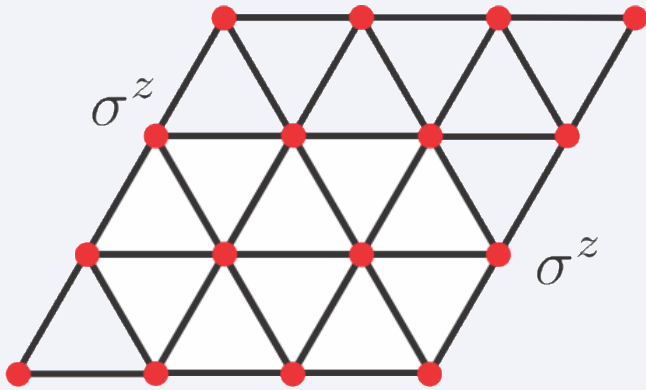
- $\mathbb{Z}_2$  SPT in (2+1)D, hosts symmetry-protected sign problem (Ellison 2021)
- Domain wall parity changes **even** number of times in SSE, net sign is 1
- Related to Ising paramagnet by  $U = (-1)^{\#DW}$
- Sign structure preserved under  $\sigma^x$  and  $\sigma^z$  perturbations.

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# Gauging the Levin-Gu model



$$H_{DS} = - \sum_j \sigma_j^x \prod_{\Delta_{jkl}} i^{\frac{1}{2}(1 - \sigma_k^z \tau_{kl}^z \sigma_l^z)} - \sum_{\Delta_{jkl}} \tau_{jk}^z \tau_{kl}^z \tau_{lj}^z - \sum_j \sigma_j^x \prod_k \tau_{jk}^x$$

$$H_{TC} = \sum_j \sigma_j^x - \sum_{\Delta_{jkl}} \tau_{jk}^z \tau_{kl}^z \tau_{lj}^z - \sum_j \sigma_j^x \prod_k \tau_{jk}^x$$

$U$  is globally symmetric and can be gauged

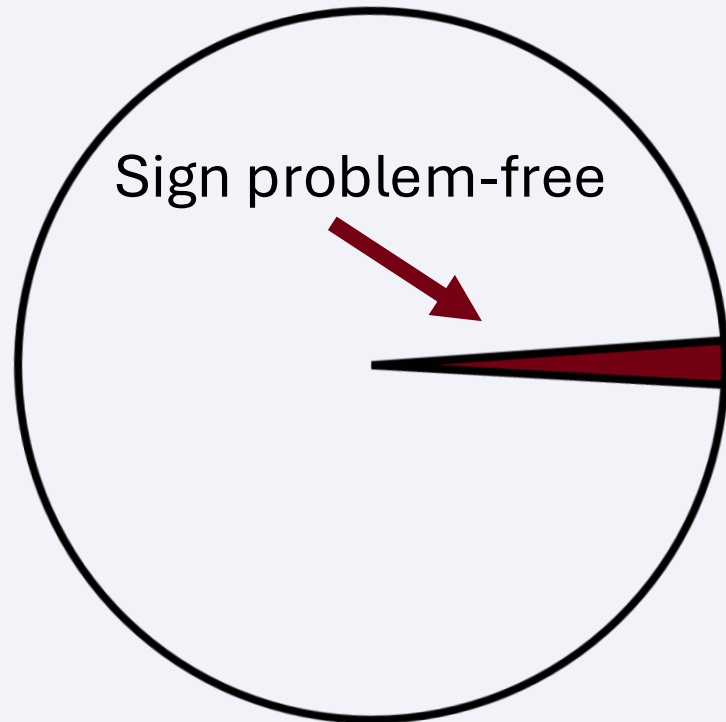
SSE sign structure of  $H_{TC}$  and  $H_{DS}$  are identical

# Generic case

- SPT Hamiltonian related to trivial stoquastic model via SPT entanglers (attaches phases according to group cohomology)
- Gauging yields generic twisted quantum double
- Can condense flux excitations, access phase transitions

# What are the true limits to simulatability?

Before:

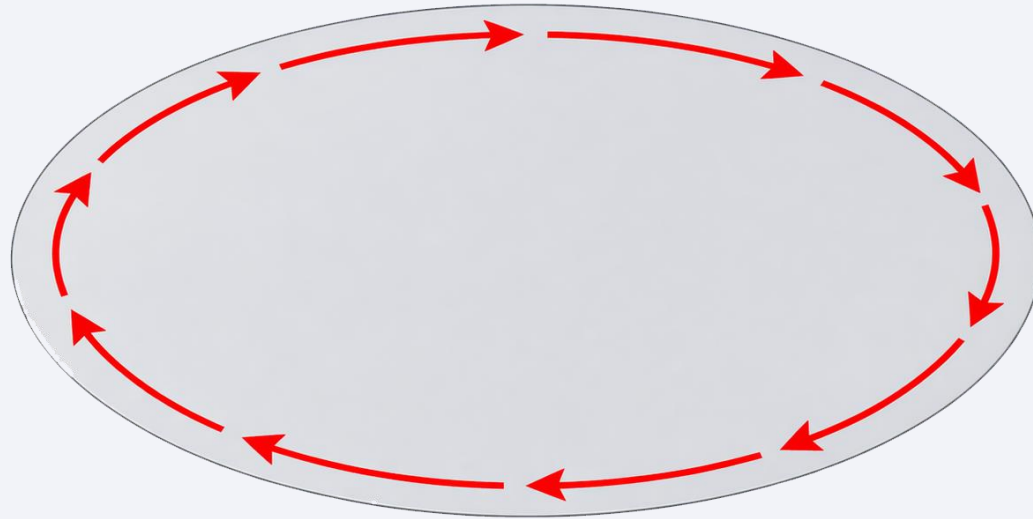


Now:



# What are the *true* limits to simulatability

- Refined criteria: sign problem-free models are *stoquasticizable*
- Are there obstacles to stoquasticizability?
- *Chiral* phases are an obstruction!



# Thank you Subir!

